

GERARDO ANTONIO CORRAL RUIZ

Applied AI/ML Research • Data Science • Master's in Digital Innovation & Governance



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PROFESSIONAL SUMMARY

Master's student at the University of Bologna with a strong focus on **ML, AI, and digital governance**. My work combines applied AI/ML research with a **critical and interdisciplinary perspective on technology, emphasizing social and environmental impact, governance, and digital transformation**. I have contributed to projects involving applied machine learning and the integration of **AI agents to enhance industrial cybersecurity processes and support strategic decision-making across the agricultural, energy, environmental, and healthcare sectors**. In addition to my technical background, I have experience with **agile methodologies and collaborative digital tools**. As a **lecturer and researcher in philosophy and contemporary culture**, I also explore the **ethical, social, and political implications of emerging technologies**, fostering critical discussion around the **relationship between technology, society, and governance**.

PROFESSIONAL EXPERIENCE

Applied AI Researcher — Weather & Energy Forecasting @ Hypermeteo (Italy)

April 2026 – Present

Conduct applied research in **AI, ML, and DL** for nowcasting and short-term forecasting of solar radiation in the **energy sector**. Develop workflows for **meteorological and satellite data analysis**, including validation, pattern and anomaly detection, and climate data interpretation. Generate actionable insights for **decision-making and perform large-scale experiments on HPC infrastructure**.

Academic Researcher (Mexico)

- **Researcher in training** — "Racionalidad, acción y vitalidad de las instituciones sociales" @ UP
- **Research Scholarship Recipient and Research Assistant** – Dr. Héctor Zagal @ UP

2025–Present
2018–2023

AI/ML Applied Research & Advisory @ Syntezia Sàrl (Switzerland)

January 2026–April 2026

Conducted **applied research in artificial intelligence and machine learning for cybersecurity in IT and OT/IIoT environments**, with a focus on **ML/DL-based intrusion detection systems using industrial telemetry and network traffic data**. Contributed to **ecosystem mapping activities involving EDIHs, SMEs, and potential strategic partners within the European digital innovation landscape**.

University and High School Lecturer (Mexico)

- **Universidad Panamericana** – Lecturer "**Selected Topics in Contemporary Culture**".
- **Northridge School Mexico** – Full-time **Spanish and Philosophy** Teacher.

2025
2023–2024

Serious Game Development & Project Coordination @ PCI Lab (México)

2020–2022

Contributed to the development of a **serious game focused on emotional intelligence for young people**, working in the **Narrative Design team**. Worked within **agile workflows with weekly sprints, managing backlog and roadmap processes**. The project received the **ILAN Innovation Award**.

EDUCATION

Master's in Digital Innovation & Governance @ University of Bologna (Italy)

2024–2026

Selected as **class representative** in the master's program. **Current GPA: 29.43/30**. Developing thesis research in collaboration with Hypermeteo in the field of **applied AI for climate and energy forecasting**.

- **Machine Learning for Cardiovascular Disease Detection (Laboratorio di Programmazione, INF-05/A, 2024)**
 - End-to-end academic data science project based on Kaggle's Heart Disease Dataset.

GRADE: 30/30

- **Moderate and Extreme Temperatures and Climate-Health Risk in Bologna: A Data-Driven Climate Risk Study (Metodi Informatici per la Trasformazione Digitale, INF-05/A, 2025)**
 - Identified high-risk days for vulnerable populations to support public policy decision-making.

GRADE: 30/30
cum laude

- **AI and the Digital Panopticon: Between Security and Freedom (*Governance e Diritto delle PPAA, IUS/09, 2024*)** **GRADE: 30/30 cum laude**
 - Analyzed the impact of AI on European public security, with particular attention to the AI Act (EU Regulation 2024/1689) and the GDPR.
- **Cybersecurity Threats in IIoT: Can ML Mitigate Vulnerabilities and Prevent Harm? (*Governance della Cybersecurity, ING-INF/05, 2025*)** **GRADE: 30/30 cum laude**
 - Developed a research paper on cyber risk and governance in IIoT/OT environments, including a Python-based case study using ToN-IIoT and Modbus telemetry. Evaluated ML models —DT, RF, and SVM— for binary and multiclass attack classification. The project explored how ML systems can support SOC analysts and strategic cybersecurity decision-making and highlight the hidden threats among an industry.

B.A. in Philosophy @ Universidad Panamericana (Mexico) **2018–2023**

- Selected for an academic exchange program at the **University of Oviedo, Spain** (2022–2023).
- **President of the Student Council**, School of Philosophy (2021–2022).

Incomplete Studies in Digital Animation and Video Game Engineering @ Universidad Panamericana (Mexico) **2017–2019**

CONFERENCES & PUBLICATIONS

- Speaker: "Emerging Risks and Threats of Digitalization and AI in Hospitals and Healthcare Services"** **2026**
— Mexican Red Cross (DIPEC)
https://www.facebook.com/dipeccruzroja?locale=es_LA
- Speaker: "Liquid Identity in the Age of Social Media"** – Instituto Antonio Nariño, **Colombia** **2024**
<https://www.filosofoteca.app/volunteer-speakers>
- Publication:** Article on **generative AI and the future of the arts** in *Generación Universitaria (El Universal)*. **2022**
<https://www.generacionuniversitaria.com.mx/tu-voz/dibujo-asistido-por-inteligencia-artificial-es-arte/>
- Moderator and Discussant:** XXIV Student Colloquium, School of Philosophy, **Universidad Panamericana**. **2020–2021**

CERTIFICATIONS, COURSES & HACKATHONS

AI, DATA SCIENCE & BUSINESS ANALYTICS

- iFAB 4 Next Generation Talents** – iFAB & Hypermeteo **2026**
- AI Automation Hackathon powered by n8n** – Jenia **2026**
- Intelligenza artificiale e business intelligence: strumenti e algoretica** – IFOA **2025**
- EITC/AI/MLP Machine Learning with Python** – European Information Technologies Certification Institute (EITCI) **2025**
- Seminario LEEIA Lógica, Epistemología y Ética de la Inteligencia Artificial** – Universidad Panamericana **2025**
- Strumenti di data analysis e visualization** – IFOA **2025**
- EITC/CP/PPF Python Programming Fundamentals** – European Information Technologies Certification Institute (EITCI) **2025**
- Business Analytics with Excel: Elementary to Advanced** – Johns Hopkins University **2023**
- What is Data Science?** – IBM **2023**
- PredictionX: Omens, Oracles & Prophecies** – HarvardX **2020**

MANAGEMENT & INNOVATION

- Management: A Professional Future** – IESE, University of Navarra, Barcelona **2023**
- Introducción a la Dirección de Empresas (IDE)** – IPADE Business School **2021**

DIGITAL ARTS, NARRATIVE COMMUNICATION & RHETORIC

- Second Place** in the "**About AI**" category and **Third Place** in the "**Made by AI**" category — ALMA-AI Video Contest **2025**
- Story and Narrative Development for Video Games & Introduction to Game Design** – California Institute of the Arts **2022**
- Modern Arts & Ideas** – MoMA Certificate **2021**
- Studies in Cinematography and Digital Post-Production** – SAE Institute **2019–2020**
- Teatro UP** – Universidad Panamericana **2017–2020**

TECHNICAL SKILLS

- **Agile Methodologies & Collaboration Tools:** Essence Workbench, Taiga, Notion, Slack, Airtable, Protégé, LaTeX, Jira.
- **Data Analysis & Programming:** Python, R, SQL, Git, HTML, CSS, JS.
- **Generative AI Tools:** ChatGPT, Gemini, NotebookLM, Sora, Midjourney, ComfyUI, Codex
- **Design & Editing Software:** Adobe (Premiere, Photoshop, After Effects), Canva, Blender
- **Microsoft Office:** Word, PowerPoint, Excel

LANGUAGES

- **Native:** Spanish
- **Advanced:** English, Italian
- **Basic:** Japanese

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